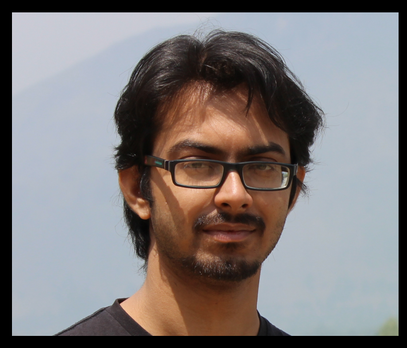
Extreme magnetic fields around black holes

Koushik Chatterjee (KC) [1,2,3,4], Nicholas Kaaz [5], Matthew Liska [6,7], Sasha Tchekhovskoy [5], Sera Markoff [8,9], Ramesh Narayan [3,4]

1: CITA, 2: UMD, 3: BHI, 4: CfA, 5: CIERA, 6: CRA, 7: ITC, 8: API, 9: GRAPPA

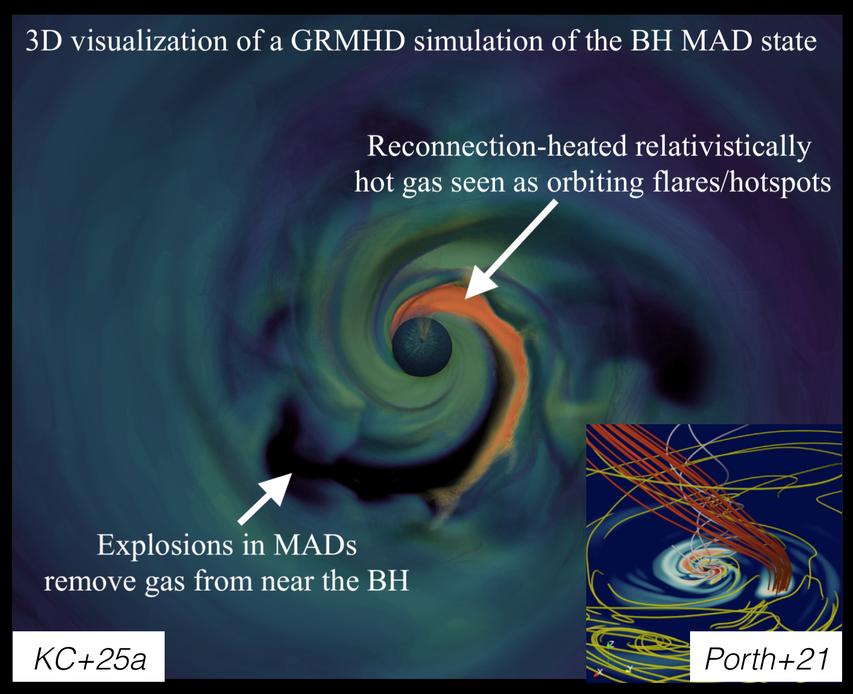


Magnetically Arrested Disks (MADs)

The black hole accumulates magnetic flux until saturation (Narayan+03, Tchekhovskoy+11, Ripperda+22)
*Over-saturation leads to frequent ejection of magnetic flux tubes into the accreting gas: **BH feedback** (Chatterjee+22)*

Eruptions drive disk winds, jet boundary waves and regulate the disk angular momentum (Chatterjee+25a)

Scan for Chatterjee+25a for a 'MAD' review

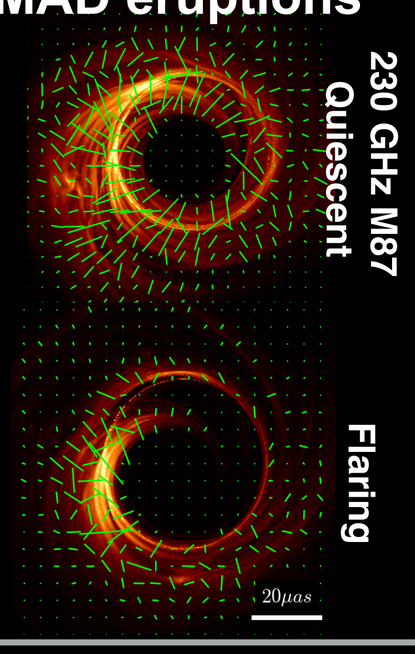


Multi-wavelength Flares from MAD eruptions

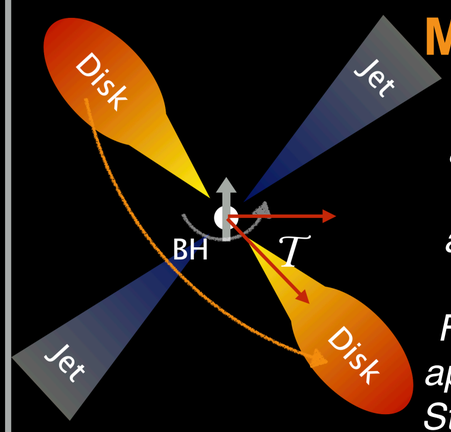
"Anti-flare" (dip in emission) in sub-mm & radio in Sgr A/M87 as relativistically-low temperature gas is pushed out (links to EHT physics: Gelles, KC+22, Jia,..,KC+23)*

Near-Infrared and X-ray flares in Sgr A & TeV flares in M87 from reconnection events around the edge of flux tubes (Dexter+20, Ripperda+22, Hakobyan+23, Solanki+25)*

Scan for a movie of a BH flare!

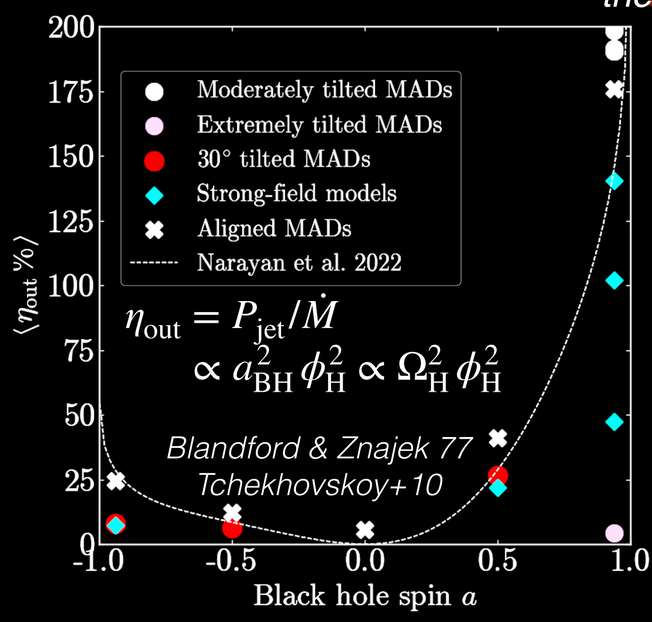


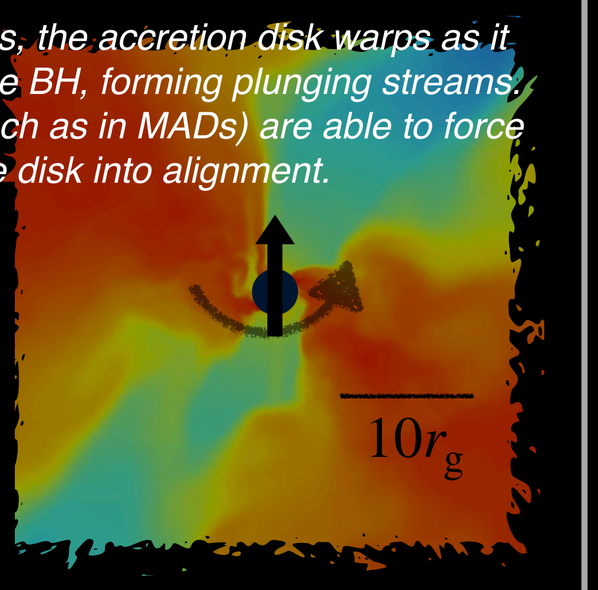
Misaligned Black hole Accretion



In highly sub-Eddington accretion disks, the accretion stays vertically-thick out to large radii (e.g., Bondi radius) but may have misaligned angular momentum with respect to the BH spin

For such flows, the accretion disk warps as it approaches the BH, forming plunging streams. Strong jets (such as in MADs) are able to force the disk into alignment.

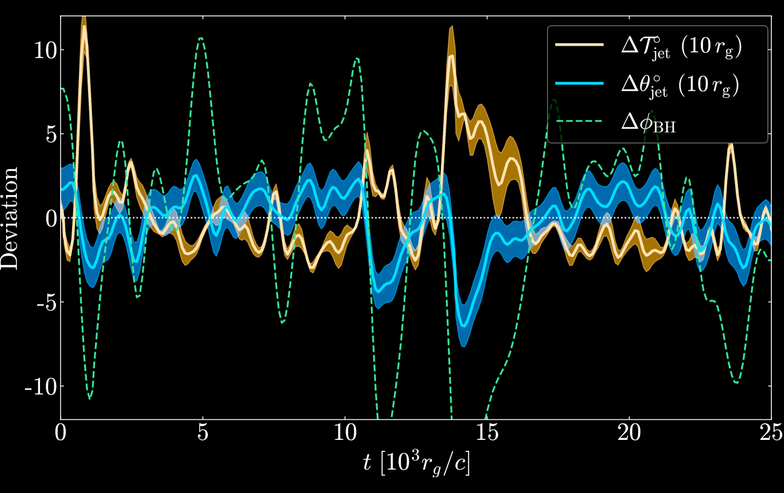




MADs produce the most powerful jets. Higher misalignment leads to more warping, and lowers the maximum jet power

Quasi-Periodicity in Jet Observations

*Magnetic flux eruptions (drops in ϕ_{BH}) change the jet's morphology: we see deviations in the jet position ($\mathcal{T}$) & the jet opening angle ($\theta_{\rm jet}$). Such oscillations in the jet base could explain **shifts in blazar jets** due to Doppler effects*

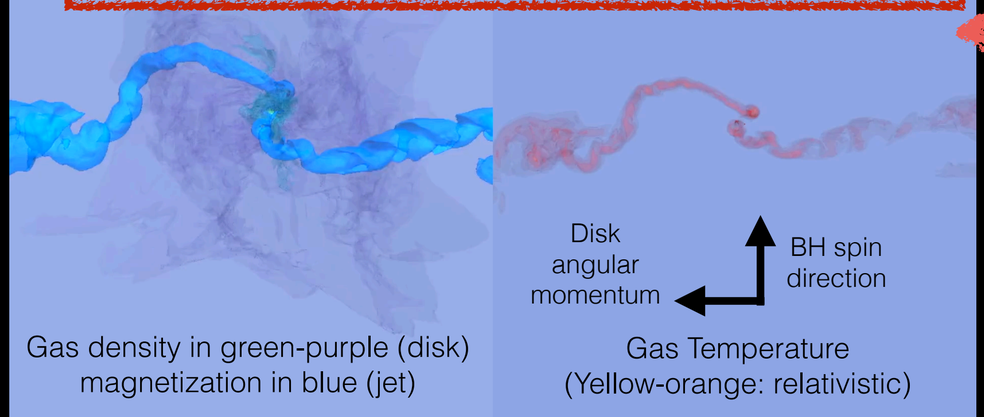


Black Hole feedback via disrupted jets

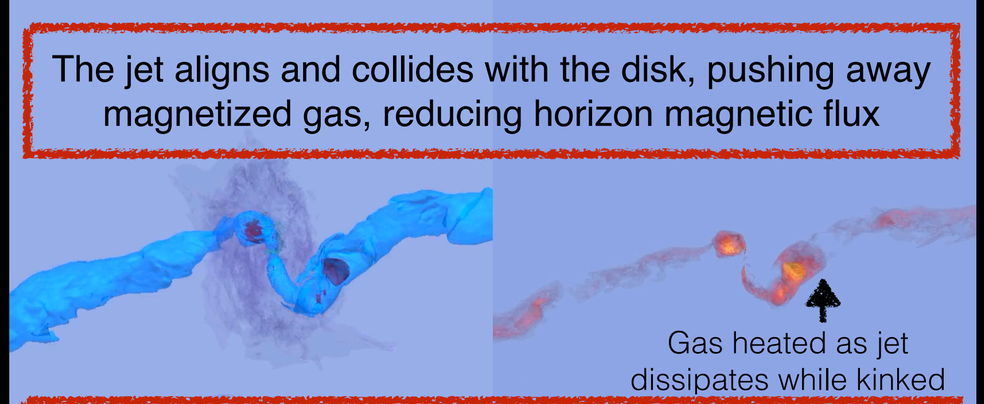
For extreme misalignments ($\mathcal{T}$>60°), the jet collides with the vertically-thick accretion flow, drives shocks and reconnection events, kinks and entrains gas, inflates cavities and eventually shreds into magnetized large-scale filaments.

Scan for Chatterjee+25b for Misaligned MADs

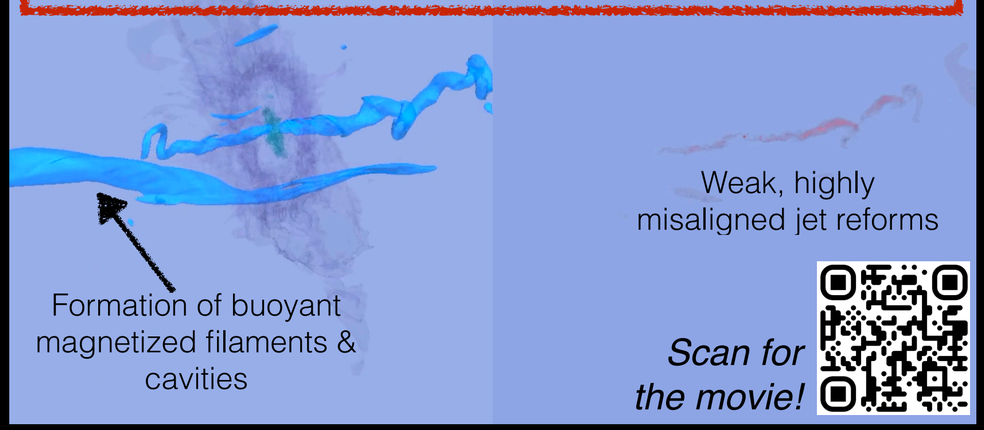
An extreme case of a 90° misalignment:
The jet launches perpendicular to the black hole spin



The jet aligns and collides with the disk, pushing away magnetized gas, reducing horizon magnetic flux

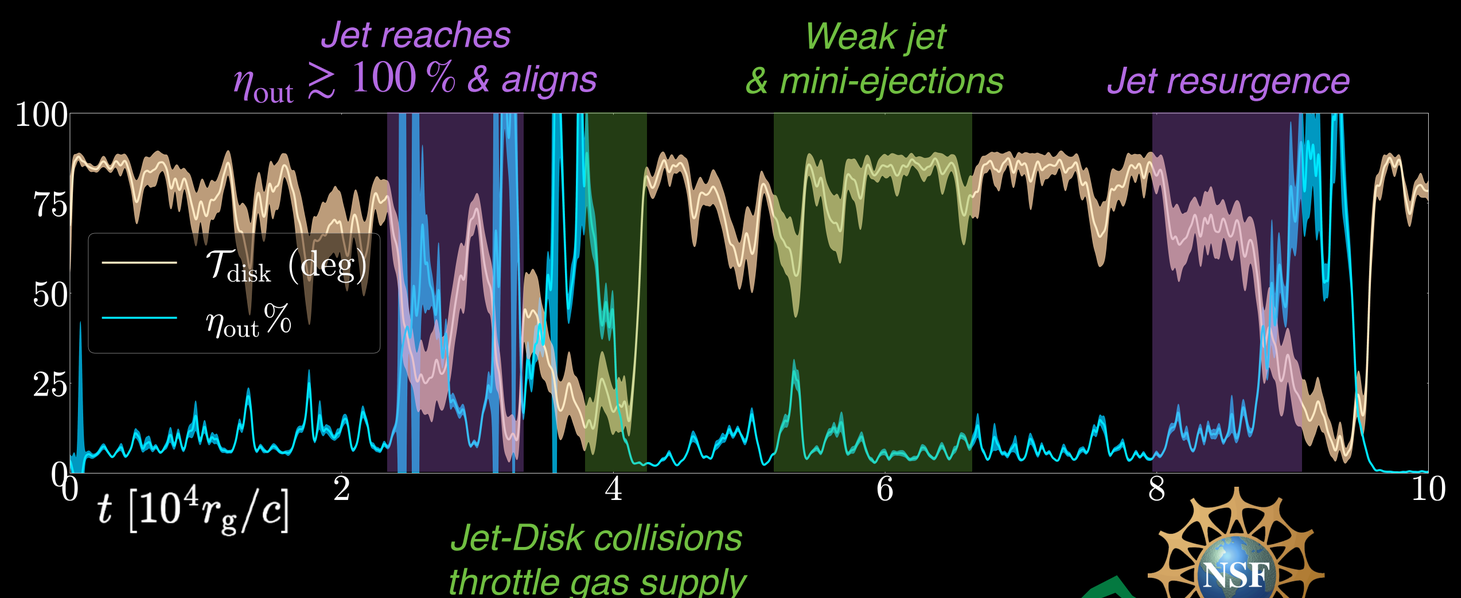


Jet quenching and slow restart of accretion:
magnetic flux re-accumulates in the ergoregion



Scan for the movie!

Filaments generally follow the disk angular momentum direction. Usually these magnetized filaments get sheared out by disk rotation.
Can they stay coherent in low-angular momentum accretion flows and explain galaxy-scale filaments? Magnetized Bondi-like flow simulations show large-scale flux tubes (Lalakos, ..KC+25, Galishnikova,..,KC+25) that could fit the bill!



Intermittent jets with a ~50,000 r_g/c period (viscous timescale at 500r_g) (~3-days to 8 yrs for $10^{6\text{-}9}$ $M_\odot$ black hole):
Candidate X-ray and Radio outbursts in otherwise radio-quiet or non-Jetted AGN

SMBHs & Blue Notes